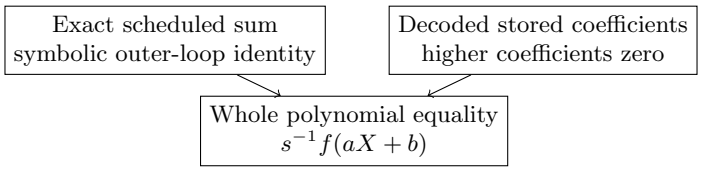
Word traversal equals affine polynomial composition 2026-10-31

T-1171. Producer proof; independent source-fidelity review pending.

The complete decoded output polynomial now satisfies

$$\text{decodedPoly}(\text{transform}(c, a, b, s)) = C(s^{-1})\, f(aX + b).$$

The proof identifies the exact ordered list of binomial contributions with the earlier symbolic outer-loop polynomial, matches every stored coefficient, and proves higher coefficients vanish. If the decoded scale satisfies $s = a^m$, the result equals the existing normalization map T_m.



The generic list/binomial identity allows arbitrary coefficient count over a field. The word-array theorem retains length ≤ 6, prime $1 < p < 2^{31}$, canonical coefficients and parameters, and nonzero scale. The scale equality in the T_m corollary is an explicit hypothesis.

Twelve main declarations and ten controls passed, with 22 named standard-axiom readbacks. Controls include empty/constant polynomials, identity at arbitrary length, a shifted/scaled quadratic, negative alpha, omitted scale, quintic leading coefficient and vanishing higher coefficients. For $f = 1 + 2X + 3X^2$ and $(a, b, s) = (2, 3, 4)$, the rational output coefficients are

$$[17/2,\ 10,\ 3].$$

Changing a to -2 changes the linear coefficient to -10; omitting the scale changes the constant coefficient. These are formal controls, not cohort measurements.

Identification attempt 1 required explicit list/function/coefficient rewrites. Attempts2–3 corrected a closed-goal tactic, notation parsed as a principal-segment token, and a conditional equality orientation. Attempt 4 passed. Reconstruction attempt 1 lacked an explicit prime-field instance; native exit 1 rejected it, including elaborator-generated sorry warnings. Attempt 2 passed; final named axiom readback contains no `sorryAx`. Controls and readback passed first attempt. All failures and nonfatal warnings are retained.

Sources: `projective_law_gate.rs` transform and the preceding loop, schedule, traversal, storage and arithmetic proofs. Primary reference: Mathlib polynomial coefficient identities, through pinned imports.

Next source bindings: scale computation for the corollary, trailing-zero trimming and fixed-size divisor copying. This logical source model is not compiled-Rust equivalence or a geometric class transport proof. Existing compiled dependencies remain in the trust boundary. No scientific campaign, timing or cost ratio.